

EXAM SECTION A

1. What is the derivative of $\cos(x)$?

(A) $-\sin(x)$

(B) $\sin(x)$

(C) $\tan(x)$

(D) $-\cos(x)$

2. Consider an ideal Carnot engine operating between temperatures T_1 and T_2

- (A) Efficiency depends only on T_1 and T_2
- (B) Efficiency is always 100%
- (C) Efficiency is independent of working substance
- (D) Both A and C

3. Define entropy in statistical mechanics.